

WeAgri

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CC 206 | Group 4



A Single Page Application (SPA)
with Real-Time Consultation
Services for Farmers

Introduction

a **Single Page Application (SPA)** designed to support farmers by providing real-time agricultural assistance through features like live chat, notifications, and dynamic updates. The platform allows farmers to consult agricultural experts about concerns related to crops, pests, soil management, and farming techniques. It also includes an AI-powered chat assistant that can instantly answer simple or common questions, while more complex concerns can be directed to human agricultural experts for further guidance.



Objectives



- To develop a **real-time consultation platform** that connects farmers with agricultural advisers for faster and more accessible guidance.
- To develop an **AI-powered chat assistant** using AgroLLM within a Retrieval-Augmented Generation (RAG) modeling approach, providing accurate, context-based answers to common farming questions while directing complex concerns to agricultural experts.
- To generate more informed and timely farming decisions that can help improve agricultural productivity and efficiency.



Methods

WeAgri was developed as an integrated web-based agricultural assistance platform that combines consultation management, AI support, farm monitoring, and communication tools in one system. The updated version uses a PHP-based application with JavaScript-driven frontend behavior, MySQL as the main database, and local JSON storage as a fallback when live database services are unavailable. The system was designed to support both consultation workflows and dashboard-based farm monitoring, making it more than a simple messaging platform.

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** PHP with API endpoints
- **Data storage:** MySQL with JSON fallback
- **User roles:** farmer, consultant, administrator
- **Modules:** consultation, dashboard, messaging, knowledge base, notifications, feedback



System Process

The updated system starts when a **farmer logs in, enters farm details, and submits a concern or asks AgroLLM for help.** The AI checks the concern against stored agricultural knowledge, identifies the crop and issue, and decides whether to provide advice or refer the case to a consultant. The system also updates dashboard data, notifications, and messages automatically.

- Farmer submits a concern or question
- AI checks the knowledge base
- System identifies crop and issue
- Concern is answered or escalated
- Admin manages assignments and status
- Dashboard and messages refresh automatically



Results

The **WeAgri** prototype successfully combines consultation services, AI guidance, dashboard monitoring, and communication tools in one platform. The system no longer focuses only on submitting concerns, because it now also supports weather viewing, market price monitoring, consultant messaging, notifications, and feedback collection. These results show that the project can assist farmers not only in reporting crop problems, but also in monitoring farm-related information and receiving timely support.

- Supports **role-based access** for farmers, consultants, and administrators
- Allows farmers to submit concerns and receive **AI-generated initial guidance**
- Refers more complex cases to human consultants
- Displays dashboard data such as temperature, soil moisture, crop condition, and open queries
- Includes market prices, notifications, and direct messaging
- Uses a **knowledge-based AI assistant** for more grounded responses



Results

The current project data shows that the system is already working as an **active prototype with stored records and actual user interactions**. These records indicate that WeAgri can already manage consultation processing, AI support, consultant communication, and monitoring features within one environment.

- The consultation workflow is functioning from submission to expert assignment
- The AI assistant now uses stored agricultural knowledge
- Both consultation threads and direct consultant chat are available
- Notifications and feedback features are already active
- The system is functional as a prototype, but still needs formal testing for full deployment

Demo

